



Harvester & Rancher Prime

Automated Bare-Metal Deployment with Ansible

Technical Documentation

PXE-based Harvester HCI installation · Edge Image Builder · Rancher Prime on RKE2

Version 1.0 · July 2026

Sanitized for public release — no credentials or personal data included

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1. Overview and Architecture

This project provides a fully automated, repeatable deployment of a SUSE Harvester hyper-converged infrastructure (HCI) cluster on bare-metal hardware, followed by a highly available Rancher Prime management cluster running as virtual machines on top of Harvester. The entire lifecycle is driven by Ansible playbooks executed on a single openSUSE Leap 15.6 administration host.

The deployment consists of three layers:

- **Admin / PXE host** — an openSUSE Leap 15.6 machine running dnsmasq (proxy DHCP), TFTP, nginx (HTTP file server), an NFS server for backups, Docker for the SUSE Edge Image Builder, and Ansible itself.
- **Harvester HCI cluster** — three bare-metal nodes installed unattended via iPXE network boot. The first node creates the cluster and claims the virtual IP (VIP); nodes two and three join automatically.
- **Rancher Prime cluster** — three RKE2 server VMs booted from a self-installing SL Micro ISO built by the SUSE Edge Image Builder. The image contains RKE2, cert-manager and Rancher Prime pre-configured, so the cluster comes up without any manual steps. A Harvester load balancer exposes Rancher on ports 80/443 through a dedicated VIP.

1.1 How the automation fits together

Node identity is anchored to MAC addresses throughout the whole pipeline. During PXE boot, dnsmasq hands the machine an iPXE binary, iPXE chain-loads a boot menu from the HTTP server, and the menu routes each machine to a node-specific script based on its MAC address. That script boots the Harvester kernel with a node-specific install configuration (create for node 1, join for nodes 2 and 3). The same principle is used for the Rancher VMs: each VM is created with a fixed MAC address, and the Edge Image Builder matches its nmstate network profile (static IP, hostname) to that MAC at first boot.

All post-installation configuration — storage classes, cluster networks, VM images, the backup target, virtual machines, load balancers, and the Rancher/Harvester integration — is applied declaratively through the Kubernetes APIs of Harvester and Rancher using the `kubernetes.core` Ansible collection. Every playbook is idempotent and can be re-run safely.

1.2 Component versions

Component	Version
Harvester	v1.7.1
Rancher Prime	2.13.3
RKE2	v1.33.6+rke2r1
cert-manager	1.19.2
Edge Image Builder	1.3.2 (SUSE Edge 3.5)
Base OS (image)	SUSE Linux Micro 6.1
Admin host OS	openSUSE Leap 15.6

2. Prerequisites

Area	Requirement
Admin host	openSUSE Leap 15.6, at least 2 vCPU, 8 GB RAM and 120 GB free disk space (the Edge Image Builder unpacks and repacks a full ISO). These limits are enforced by assertions in the <code>eib_common</code> role.

Area	Requirement
Harvester nodes	Three bare-metal servers with PXE-capable NICs. The MAC address of the management NIC of every node must be known and entered in <code>vars/infrastructure.yml</code> .
Network	Admin host and nodes share one Layer-2 segment. Free static IP addresses are required for all nodes plus two VIPs (Harvester GUI/API and Rancher).
Ansible	Version 2.14 or newer with the <code>community.general</code> and <code>kubernetes.core</code> collections (install via <code>requirements.yml</code>).
Licensing	A SUSE Customer Center registration code for SUSE Linux Micro and access to the Rancher Prime chart repository.
TLS	A server certificate, private key and root CA for the Rancher FQDN, placed in the <code>cert/</code> directory (see <code>cert/README.md</code>).

```
ansible-galaxy collection install -r requirements.yml
```

3. Security and Data Sanitization

This repository has been prepared for public release on GitHub. All personal and security-relevant data contained in the original project has been removed and replaced by neutral placeholders or Ansible variables:

Removed / replaced	How it is handled now
SSH public keys (personal)	Variable <code>ssh_public_key</code> in <code>vars/infrastructure.yml</code> (<code>CHANGE_ME</code> placeholder), referenced by all templates and the <code>harvester_ssh</code> role.
Cluster tokens and passwords	Placeholders (<code>CHANGE_ME</code>) in <code>vars/infrastructure.yml</code> ; the documentation recommends Ansible Vault for real values.
Encrypted OS password hashes	Variable <code>node_encrypted_password</code> ; generate your own hash with <code>"openssl passwd -6"</code> .
SCC registration code	Placeholder <code>sccRegistrationCode: CHANGE_ME</code> .
Kubeconfig tokens and CA data	Real kubeconfigs are Git-ignored; sanitized <code>*.example</code> files document the expected structure.
Personal user names and hostnames	Replaced by the <code>admin_user</code> variable and the neutral <code>example.lan</code> domain.
Private network details	Example addressing (192.168.100.0/24) driven entirely by the <code>network_prefix</code> variable.
TLS certificates and keys	Never stored in Git; the <code>cert/</code> directory is excluded via <code>.gitignore</code> .

3.1 Recommended practice

- Encrypt every secret with Ansible Vault before committing anything: `ansible-vault encrypt_string 'MySecret' --name 'harvester_cluster_token'`.
- Never remove the `.gitignore` entries for kubeconfigs, certificates and build artifacts.
- Rotate the Harvester and RKE2 cluster tokens as well as the Rancher bootstrap password after the initial deployment.
- Search the repository for the string `CHANGE_ME` before the first run — every occurrence marks a value that must be adapted.

4. Configuration Reference

All environment-specific values are concentrated in the `vars/` directory. No IP address, credential or hostname is hard-coded anywhere else.

4.1 vars/network.yml

Variable	Purpose
<code>network_prefix</code>	First three octets of the management network; almost every address is derived from it.
<code>pxe_ip</code>	IP of the admin host — acts as PXE, HTTP, TFTP and NFS server.
<code>dhcp_range</code>	Temporary DHCP range used only during PXE boot.
<code>gateway / dns / domain</code>	Default gateway, DNS server and DNS domain for all nodes.
<code>vlan_id_harvester</code>	"0" = untagged network; any value > 0 switches all templates to their VLAN variants.
<code>harvester_vip / rancher_prime_vip</code>	Virtual IPs of the Harvester cluster and the Rancher load balancer.
<code>harvester_nodeX_ip / rancher_nodeX_ip</code>	Static addresses of the three Harvester nodes and the three Rancher VMs.
<code>vlangs</code>	List of VM networks (NetworkAttachmentDefinitions) created on Harvester.

4.2 vars/infrastructure.yml

Variable	Purpose
<code>harvester_version</code>	Harvester release; also selects the correct iPXE VLAN syntax (changed in 1.7).
<code>harvester_nodeX / _mac1</code>	Hostname and management-NIC MAC address of each bare-metal node.
<code>harvester_cluster_token</code>	Shared join token of the Harvester cluster (secret, CHANGE_ME).
<code>ssh_public_key / ssh_keypair_name</code>	Admin SSH key authorized on all nodes and registered as a Harvester KeyPair.
<code>rke2_hostname</code>	Rancher FQDN — must resolve to the Rancher VIP.
<code>rke2_cluster_token / bootstrapPassword</code>	RKE2 join token and initial Rancher admin password (secrets, CHANGE_ME).
<code>node_encrypted_password / admin_user</code>	SHA-512 password hash and name of the local user baked into the SL Micro image.
<code>rancher_nodeX / _mac</code>	Hostnames and fixed MAC addresses of the three Rancher VMs.
<code>baseimage / newimage</code>	SL Micro source ISO and the name of the ISO produced by the Edge Image Builder.
<code>download_server</code>	Internal HTTP mirror that hosts the large ISO/qcow2 source files.
<code>cert_path / key_path / rootca_path</code>	Paths to the TLS material used for the Rancher ingress.
<code>sccRegistrationCode</code>	SUSE Customer Center code for SL Micro (secret, CHANGE_ME).

4.3 vars/general.yml

Contains the package list for the admin host, the directory layout (TFTP root, HTTP share, kubectl paths), the Harvester add-on switches and the `image_list` — the set of images uploaded to Harvester in playbook 12. These values rarely need to be changed.

5. Deployment Workflow

The playbooks are numbered in execution order. Two manual checkpoints exist: powering on the bare-metal nodes after the PXE validation, and downloading the kubeconfigs from the Harvester and Rancher UIs.

Step	Playbook	What it does
01	01-playbook-prepare.yaml	Installs all packages, configures dnsmasq/TFTP/nginx, downloads the Harvester boot artifacts and the SL Micro base image, renders all iPXE scripts and node configs, installs kubectl.
02	02-playbook-edge-image-builder.yaml	Validates host resources, renders the complete EIB build tree (definition, per-node network profiles, RKE2 configs, Helm values, TLS manifests) and builds the self-installing Rancher ISO in a container.
04	04-playbook-nfs-server.yaml	Exports /srv/harvester-backups via NFS as the future Harvester backup target.
11	11-playbook-pxe-onsite-validate-setup.yaml	Starts the PXE services and validates packages, services, boot files, node configs, firewall and HTTP reachability. Report-only.
—	Power on the nodes	Node 1 installs in create mode and claims the VIP; nodes 2 and 3 install in join mode. Afterwards, download the kubeconfig from the Harvester UI and save it as ./harvester_kubeconfig.yaml.
12	12-playbook-harvester-config.yaml	Stops the PXE services, registers the SSH key, sets the NFS backup target, enables auto disk provisioning, tags the data disks and creates the default "ssd" storage class, uploads the VM images, creates the cluster networks, the dedicated storage network and the VM VLANs.
13	13-playbook-rancher.yaml	Creates the DVD and boot volumes, the three Rancher VMs (fixed MACs), the IP pool holding the Rancher VIP and the load balancer for ports 80/443.
15	15-playbook-rancher-config.yaml	Adds the UI plugin chart repositories, installs the Harvester UI extension and imports the Harvester cluster into Rancher via the cluster-registration-url setting. Requires ./rancher_kubeconfig.yaml.

5.1 Command sequence

```

ansible-playbook 01-playbook-prepare.yaml
ansible-playbook 02-playbook-edge-image-builder.yaml
ansible-playbook 04-playbook-nfs-server.yaml
ansible-playbook 11-playbook-pxe-onsite-validate-setup.yaml
# -> power on the Harvester nodes and wait for the installation
# -> save the Harvester kubeconfig as ./harvester_kubeconfig.yaml
ansible-playbook 12-playbook-harvester-config.yaml
ansible-playbook 13-playbook-rancher.yaml
# -> save the Rancher kubeconfig as ./rancher_kubeconfig.yaml
ansible-playbook 15-playbook-rancher-config.yaml

```

6. Roles Reference

6.1 PXE server roles

Role	Responsibility
pxe_common	Installs all packages, the Python kubernetes module and the kubernetes.core collection; enables Docker.
pxe_setup	Creates the directory layout, copies the iPXE binaries (BIOS and UEFI) into the TFTP root and renders the dnsmasq proxy-DHCP/TFTP configuration.
pxe_harvester_download	Downloads the Harvester ISO, kernel, initrd and rootfs (idempotent, skips existing files).
pxe_harvester_init	Renders the boot menu, the per-node iPXE scripts and the create/join install configs. Template selection depends on the Harvester version and the VLAN setting.
pxe_onsite_on / pxe_onsite_off	Starts respectively stops the PXE services (dnsmasq, tftp, nginx) so no stray DHCP answers remain on the network after installation.
kubect1	Installs the latest stable kubect1 with SHA-256 checksum verification.

6.2 Edge Image Builder roles

Role	Responsibility
eib_common	Asserts disk/RAM/CPU minimums, creates the build tree, renders definition.yaml and the custom first-boot script, copies the base ISO.
eib_network	Renders one nmstate profile per Rancher node; EIB matches it to the VM by MAC address to apply the static IP.
eib_kubernetes	Renders the RKE2 server/agent configs, the Helm repositories and values for cert-manager and Rancher, and the two TLS secrets (private CA and ingress certificate).

6.3 Harvester configuration roles

Role	Responsibility
harvester_ssh	Registers the admin SSH public key as a Harvester KeyPair.
harvester_settings_nfs	Sets the NFS backup target to the export on the admin host.
harvester_settings_auto_disk	Enables automatic Longhorn provisioning of all /dev/sd* devices.
harvester_storage_class	Tags the auto-provisioned disks with "ssd" and creates the new default storage class bound to that tag.
harvester_image	Uploads a VM image from the local HTTP share; skips images that already exist.
harvester_cluster_network	Creates the "stg" (storage) and "public-vm" cluster networks including their uplink configurations.
harvester_settings_storage_network	Moves Longhorn replication traffic to the dedicated storage network.
harvester_vlan	Creates a Multus NetworkAttachmentDefinition per entry of the vlans list.
harvester_vm_disk_boot / _data	Create the ISO (DVD) volume and the OS disk of a Rancher VM as block PVCs.
harvester_vm_linux	Creates the KubeVirt VirtualMachine with fixed MAC, boot order and the load-balancer backend label.
harvester_ip_pool / harvester_load_balancer	Create the IP pool holding the Rancher VIP and the L4 load balancer for ports 80/443.

6.4 Rancher configuration roles

Role	Responsibility
rancher_git_repo	Adds the rancher and partner-extensions ClusterRepo chart sources.
rancher_harvester_plugin	Installs the Harvester UI extension (Virtualization Management).
harvester_settings_cluster_registration_url	Creates the Rancher cluster object, waits for the registration token, builds the import URL and writes it into the Harvester settings — completing the integration.

7. Template Reference

Template	Rendered as	Purpose
dnsmasq-pxe.conf.j2	/etc/dnsmasq.d/pxe.conf	Proxy DHCP + TFTP; two-stage boot (PXE firmware -> iPXE -> HTTP boot menu).
boot.ipxe.j2	boot.ipxe	Main menu; routes each machine to its node script by MAC address.
ipxe-harvester*.j2	node-<mac>.ipxe	Per-node kernel boot; three variants: no VLAN, VLAN <1.7, VLAN >=1.7.
harvester-create*.j2 / -join*.j2	config-node-<mac>.yaml	Unattended Harvester install configs (create for node 1, join for the rest), with and without VLAN tagging.
definition.j2	eib/definition.yaml	Main Edge Image Builder definition: OS users, packages, RKE2 topology, Helm charts.
hostname.j2	eib/network/<fqdn>	nmstate profile per Rancher VM (static IP, matched by MAC).
helm.j2 / rancher.j2 / certmanager.j2	eib/kubernetes/helm/values/*	Helm repositories and chart values for Rancher Prime and cert-manager.
rke2_server.j2 / rke2_agent.j2	eib/kubernetes/config/*	RKE2 token and TLS-SAN configuration.
tls-secret.j2 / tls-rancher-ingress.j2	eib/kubernetes/manifests/*	Private CA and ingress certificate as Kubernetes secrets.

8. Troubleshooting

Symptom	Resolution
A node does not PXE boot	Run playbook 11 for a full status report. Check "journalctl -u dnsmasq -f" on the admin host while the node boots and verify its MAC address in vars/infrastructure.yml.
The Harvester installer hangs	Open the node console. The config URL printed by iPXE must be reachable; test with "curl http://<pxe_ip>/harvester/boot.ipxe".
Image upload stays at 0 %	The VirtualMachineImage URL points at the admin host — nginx must still be running (role pxe_onsite_on).
Rancher VMs get no IP address	The VM MAC addresses (playbook 13) must exactly match the MACs in the EIB network profiles (playbook 02).
Playbook 13 fails on the image ID	vm_image_id and vm_disk0_storage_class must match the names Harvester generated after the image upload; verify them in the Harvester UI under Images.

Symptom	Resolution
Cluster import does not appear	Playbook 15 needs both kubeconfigs. Check the cluster-registration-url setting: <code>kubectrl get settings -n harvester</code> .

Tip: all Harvester settings can be inspected at any time with `"kubectrl get settings -n harvester --kubeconfig ./harvester_kubeconfig.yaml"`.